

Smaran Rangarajan Bharadwaj

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Python3 • Go • C • Java • JavaScript

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EDUCATION

- **RV University, Bengaluru** · B.Tech (Hons.), Computer Science and Engineering Graduating: Aug 2026 · CGPA: 9.80
- **Sri Kumaran Children's Home, Bengaluru** · Class 12 (CBSE) - PCMC 2022 · Score: 95.2%
- **Sri Kumaran Children's Academy, Bengaluru** · Class 10 (CBSE) 2020 · Score: 90.0%

TECHNICAL SKILLS

- Developer Tools:** Docker, Kubernetes, Git/GitHub, Amazon Web Services (AWS), Google Cloud Platform (GCP), Azure, Jira, Confluence, HuggingFace
- Databases:** MySQL, SQLite, MongoDB, ClickHouse
- Frameworks:** TensorFlow, PyTorch, YOLO, Flask, Streamlit, Ollama, Apache Kafka, Apache Flink, Unsloth, gRPC, REST-API
- AI Coding Assistants:** Claude, Cursor, Devin, GitHub Copilot, Gemini

EXPERIENCE

Software Development Intern Jan 2026 - Jul 2026
Aviatrix Systems Pvt. Ltd. Bengaluru

- Architected and shipped to production a multi-cloud risk assessment pipeline (Flink, ClickHouse) analyzing 22 resource types in under 100 seconds.
- Designed and shipped the Data-Bridge service (Golang, gRPC, ClickHouse materialized views) delivering sub-second performance for 100,000+ records.
- Automated the CI/CD deployment pipeline, reducing release time for multi-hour builds and deployments to 15-18 minutes (~68% reduction).

ML Project Intern Aug 2025 - Jan 2026
Aeronautical Development Establishment (DRDO) Bengaluru

- Architected a Mission Debriefing Expert System processing 10,000+ flight parameters across 6 missions in under 15 minutes, by building an automated multi-stage pipeline with parallelized processing.
- Engineered an event detection engine using Allen's Interval Algebra and expert systems logic to classify overlapping continuous and discrete parameter anomalies.

ML Engineer Intern Jun 2025 - Aug 2025
GovernAIce Berkeley (Remote)

- Designed an end-to-end multi-lingual RAG pipeline reducing compliance report generation time by 73% over manual processes by leveraging DeepSeek-R1:70B and MongoDB Atlas Vector Search.
- Developed a policy ingestion pipeline supporting 6 languages and 12,000+ legal documents utilizing IBM Docling and XLM-RoBERTa embeddings.

AI and Advanced Computing Intern Jun 2024 - Aug 2024
QpiAI Pvt. Ltd. Bengaluru

- Built a real-time 3D Bloch Sphere visualization tool using Three.js and contributed to a 5-qubit quantum circuit simulator, deployed onto QpiAI's QVidya.

AI/ML Intern Jul 2023 - May 2024
CarViz Paris

- Trained a ResNet-50-based tyre wear classification model achieving 93.7% accuracy on a self-curated dataset of 4,000+ real-world vehicular images.

LEADERSHIP AND AWARDS

- Founded and established RV University's IEEE Student Branch, growing it from 5 founding ExeCom members to 60+ members across 3 society chapters and 20+ events.
- Led operations across major hackathons - SIH 2022 Finals, ArgoNyx, FOSS Hack 3.0 - spanning digital design, evaluation, and infrastructure, drawing 1,700+ participants.
- Awarded merit-based university scholarships in 3 consecutive years, ranking 2nd among the university's top academic performers.
- Won 2nd place among 117 teams at the HPCC Safe Heavens Hackathon for developing a distributed computing solution for refugee support systems.

- Ranked in the top 1% of 4,000+ students and professionals nationally in NPTEL's Entrepreneurship and IP Mindset course.
- Filed and published a patent (App. No. 202341068937) for Gelos, a fitness-focused interactive game, developed collaboratively with Dr. Phani Kumar Pullela and team.
- Elected Captain and Vice Captain of Emerald House, representing the house across school-level events and mentoring fellow students.

PROJECTS

Efficient and Interpretable Deep Learning Framework for PCOS Detection *Neural Architecture Search, XAI*

- Architected a DARTS-based Neural Architecture Search pipeline discovering an optimized CNN for PCOS ultrasound classification across 1,924 images, independently achieving 97.91% test accuracy.
- Integrated Explainable AI, zero-shot, and few-shot learning methods, retaining 94.72% accuracy in few-shot, limited-data scenarios.

Braille GPT *Multi-agent System, LoRA*

- Led the development of a multi-agent system built on LoRA-fine-tuned LLaMA 3.1 8B with a Streamlit-based front end, converting text and image inputs into bilingual Braille with a 0.76 average BLEU score across translation agents.

ESP32 Flow Meter Fault Detection *TinyML, Mosquitto (MQTT), FreeRTOS*

- Engineered a fault diagnosis system for ultrasonic flow meters, independently training a lightweight DL model to classify faults into 4 states, achieving 91.67% test accuracy.
- Deployed the optimized model on an ESP32 via TensorFlow Lite and EloquentTinyML, achieving 300–400ms inference latency.

MITM Attack Simulator *Network Security, Flask, VMWare Fusion*

- Engineered a full-stack network security toolkit using Scapy to simulate MITM attacks across a multi-VM bridged network, featuring a Flask-based dashboard, processing 15,000+ packets per session.

Obstacle Avoidance for Visually Impaired *YOLOv8*

- Led a four-member team to develop a YOLOv8n-based obstacle detection system for the visually impaired, achieving 82.4% mAP on a self-collected dataset of 500+ pedestrian images.
- Built and integrated a real-time audio alert pipeline, achieving 132ms average inference latency when deployed on edge.

COURSES

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| <ul style="list-style-type: none">• Entrepreneurship and IP Mindset (NPTEL) - Top 1% Nationally• Distributed Systems (NPTEL)• Multi AI Agent Systems with CrewAI | <ul style="list-style-type: none">• Advanced Graph Theory• Google Cloud Computing Foundations• AWS Networking Masterclass - Amazon VPC Hybrid Cloud |
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